Tutorial - 11 classification

Dataset

sample	Feature1	Feature2	Feature3	Actual class	Pred prob
S_1	25	40	650	1	0.90
S_2	30	60	700	1	0.75
S_3	45	35	600	0	0.60
S_4	50	30	580	0	0.40
S_5	28	55	720	1	0.30

① Threshold value = 0.5 and compute the confusion matrix.

② Calculate the following matrics

Accuracy

precision

Recall (TPR)

F_1-score.

⇒ Based on the threshold, we will calculate the pred output

Actual Class	Pred prob	Pred output
1	0.90	1 (TP)
1	0.75	1 (TP)
0	0.60	1 (FP)
0	0.40	0 (TN)
1	0.30	0 (FN)

Actual = pred to true positive

class 1 = +ve
class 0 = -ve

The shold value = 0.5

connected with probability

prob ≥ 0.5 -> class1
prob < 0.5 -> class0

TP = True +ve
FP = Flase +ve
TN = True -ve
FN = False -ve

if donot get any one of the value then it is taken as '0'

confusion matrix

prediction 1	TP 2	FP 1
prediction 0	FN 1	TN 1
	1	0

Actual

-> metrices

* precision = $\frac{TP}{TP+FP}$ = 0.66 ≈ 0.67

* Recall (TPR) = $\frac{TP}{TP+FN}$

= 0.67

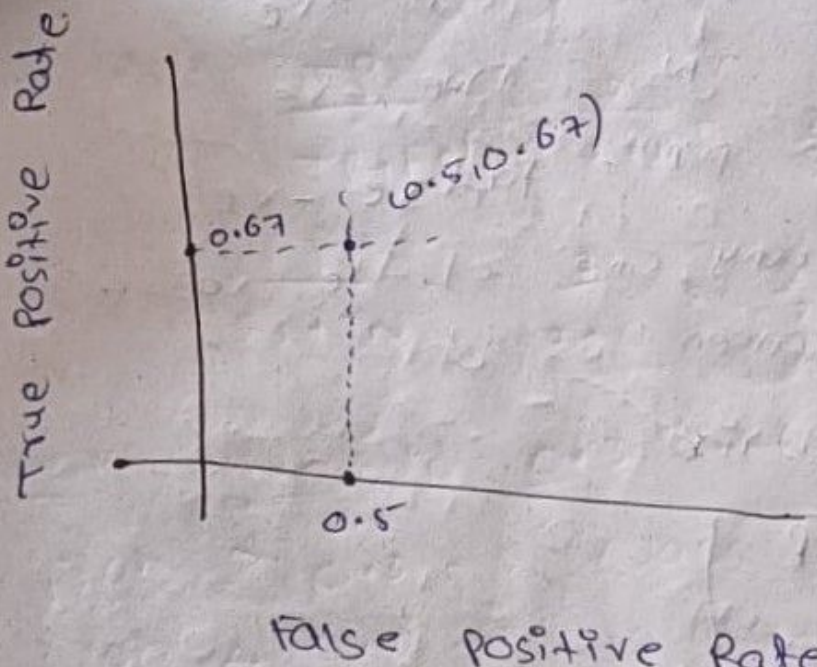
* F_1 score $= \dfrac{2 * \text{precision}}{\text{precision} + \text{Recall}}$

$= 0.67$

③ Calculate the FPR (False +ve Rate)

$FPR = \dfrac{FP}{FP+TN} = 0.5$

⑤ Plot the ROC points / curve.



④ For Threshold => 0.7, 0.6, 0.5, 0.4, 0.3

=> Finding TPR and FPR

-> For Threshold 0.7

Prob $\geq$ 0.7 -> Class -> 1

prob < 0.7 -> Class -> 0

Actual	pred	pred output
1	0.90	1 (TP)
1	0.75	1 (TP)
0	0.60	0 (TN)
0	0.40	0 (TN)
1	0.30	0 (FN)

confusion matrix.

Prediction \ Actual	1	0
1	TP 2	FP 0
0	FN 1	TN 2

$\rightarrow TPR = \frac{TP}{TP+FN} = \frac{2}{2+1} = \frac{2}{3} = 0.67$

$\rightarrow FPR = \frac{FP}{FP+TN} = \frac{0}{0+2} = 0$

$\Rightarrow$ For Threshold 0.6

prob $\geq 0.6 \rightarrow$ class - 1

prob $< 0.6 \rightarrow$ class - 0

Actual	prob	prob output
1	0.90	1 (TP)
1	0.75	1 (TP)
0	0.60	1 (FP)
0	0.40	0 (TN)
1	0.30	0 (FN)

confusion matrix.

Prediction \ Actual	1	0
1	TP 2	FP 1
0	FN 1	TN 1

$TPR = \frac{TP}{TP+FN} = \frac{2}{2+1} = \frac{2}{3} = 0.67$

$FPR = \frac{FP}{FP+TN} = \frac{1}{1+1} = \frac{1}{2} = 0.5$

→ For threshold 0.4

prob ≥ 0.4 → class – 1

prob < 0.4 → class – 0

Actual	prob	prob output
1	0.90	1 (TP)
1	0.75	1 (TP)
0	0.60	1 (FP)
0	0.40	1 (FP)
1	0.30	0 (FN)

confusion matrix

Prediction \ Actual	1	0
1	TP 2	FP 2
0	FN 1	TN 0

$$TPR = \frac{TP}{TP+FN} = \frac{2}{2+1} = \frac{2}{3} = 0.67.$$

$$FPR = \frac{FP}{FP+TN} = \frac{2}{2+0} = \frac{2}{2} = 1$$

→ For threshold 0.3

prob ≥ 0.3 → Class – 1

prob < 0.3 → class – 0.

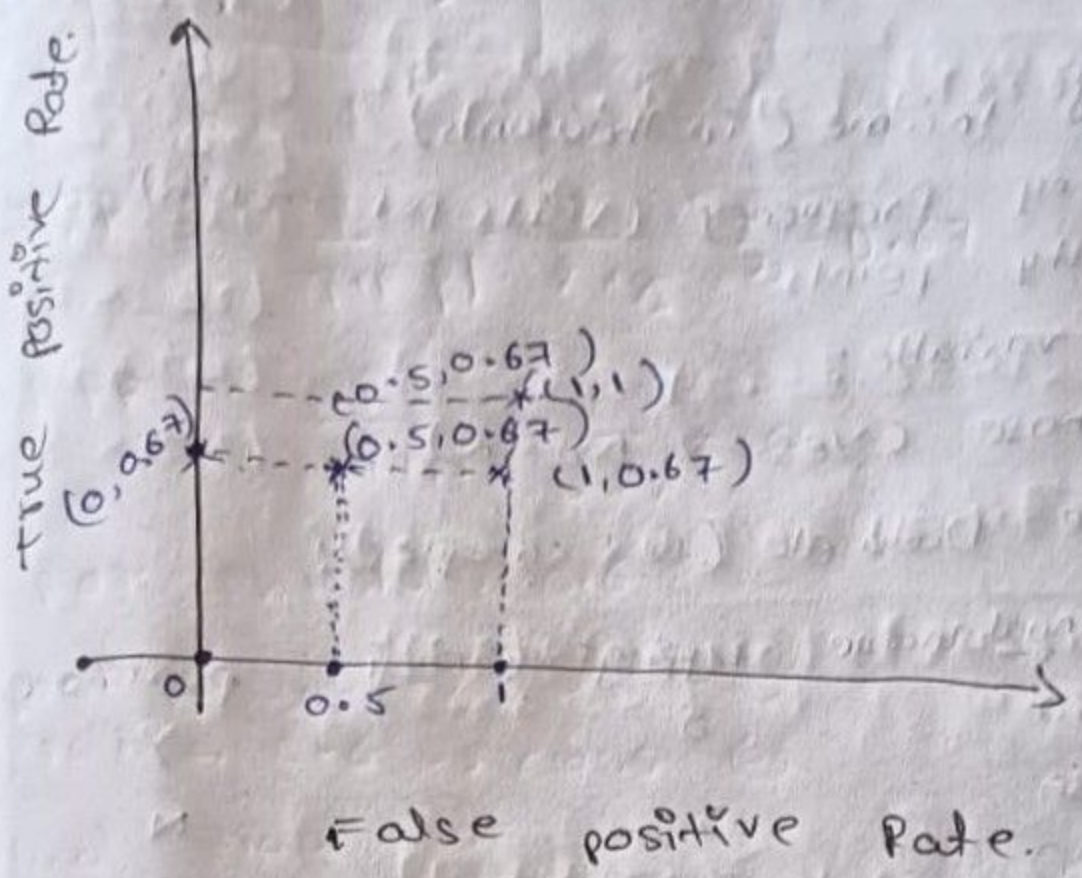
Actual	Prob	prob output
1	0.90	1 (TP)
1	0.75	1 (TP)
0	0.60	1 (FP)
0	0.40	1 (FP)
1	0.30	1 (TP)

confusion matrix

Prediction \ Actual	1	0
1	TP 3	FP 2
0	FN 0	TN 0.

$$TPR = \frac{TP}{TP+FN} = \frac{3}{3+0} = \frac{3}{3} = 1$$

$\Rightarrow FPR = \frac{FP}{FP+TN} = \frac{2}{2+0} = \frac{2}{2} = 1$

$\Rightarrow$ ROC curve for all thresholds.

$\Rightarrow$ 0.7, 0.5, 0.6, 0.4, 0.3